Lab Experiment 2: DC Motor Modelling with Load

Learning Objectives

- Derive a mathematical model of a DC motor using its electrical and mechanical equations (first principles).
- Derive an approximate linear model using input-output measured data, known as system identification, via a bump test (similar to step response modelling).
- Demonstrate competency in using MATLAB and Simulink to simulate a DC motor.
- Validate DC motor physical (electromechanical) and identified dynamic models.

Required Tools and Videos (available on Teams)

The required files and guidance videos are available in Laboratory channel **Files** tab and **Lab Guidance Videos** in MS Teams ELEC3114 class.

- DC motor parameter file
  - **ELEC3114_Lab2_New.m**
  - **ELEC3114_Lab2_QuanserDCMotorSRV02_Template_2026.slx**
- **ELEC3114 Lab 2 guidance video (coming soon!)**
- Supplementary MATLAB and Simulink familiarisation videos:
  - Simulink tutorial (Video 7 series); Transfer functions (Video 2 series); Time response of LTI systems and Stability (Video 3 series).
- **Quanser Rotary Servo Base Unit SRV02 with a disc**.

Pre-Lab 2 Exercises

NOTE: This Pre-Lab includes two MATLAB Grader exercises: Part 1 focuses on symbolic derivation of transfer functions and equations, while Part 2 covers their numerical validation. Together, they address all Pre-Lab 2 questions except Question 8, which tests conceptual understanding.

Consider the schematic diagram of the **permanent magnet brushed DC motor** used in the **Quanser Rotary Servo Base Unit SRV02** with a **load disc** as shown in Fig. 2-1. Detailed schematic model shown in Fig. 2-2. Parameters are provided in Table 2-1, **ELEC3114_Lab2.m** file and MATLAB Grader code.

1. **Derive** the DC motor **angular position** and **velocity (or speed) transfer functions**, $\theta_{ld}(s)/V_m(s)$ and $\omega_{ld}(s)/V_m(s)$ parametrically as given in Eq. 2-1(a) and Eq. 2-1(b), through the following step-by-step tasks.
   a. **Find** KVL equation describing the electrical dynamics of the DC motor in time domain, incorporating back EMF equation in terms of motor shaft angular speed $\omega_m = d\theta_m/dt = \dot{\theta}_m$).
   b. **Write down** Torque equation.
   c. **Obtain** equation of motion of the mechanical dynamics as seen from the motor shaft in time domain (i.e., in terms of motor shaft angular speed $\dot{\theta}_m$ and acceleration $\ddot{\theta}_m$).
   d. **Combine** all these equations and take Laplace Transform to calculate the transfer functions in terms of **load disc** position $\theta_{ld}(s)$ and velocity $\omega_{ld}(s)$.

   NOTE: Apply the **reflection rule** to find the total equivalent inertia J_{meq} and viscous damping coefficient B_{meq} at the motor shaft (primary side N_1). Verify using MATLAB Grader, then use gear ratios to relate motor shaft position and velocity to the load disc and derive the transfer functions.

$$\frac{\theta_{ld}(s)}{V_m(s)} = \frac{\frac{N_1 N_3}{N_2 N_4} K_m}{s\big((J_{meq}s + B_{meq})(L_m s + R_m) + K_m K_b\big)}, \quad \text{(a)}$$

Eq. 2-1

$$\frac{\omega_{ld}(s)}{V_m(s)} = \frac{\frac{N_1 N_3}{N_2 N_4} K_m}{(J_{meq}s + B_{meq})(L_m s + R_m) + K_m K_b}. \quad \text{(b)}$$

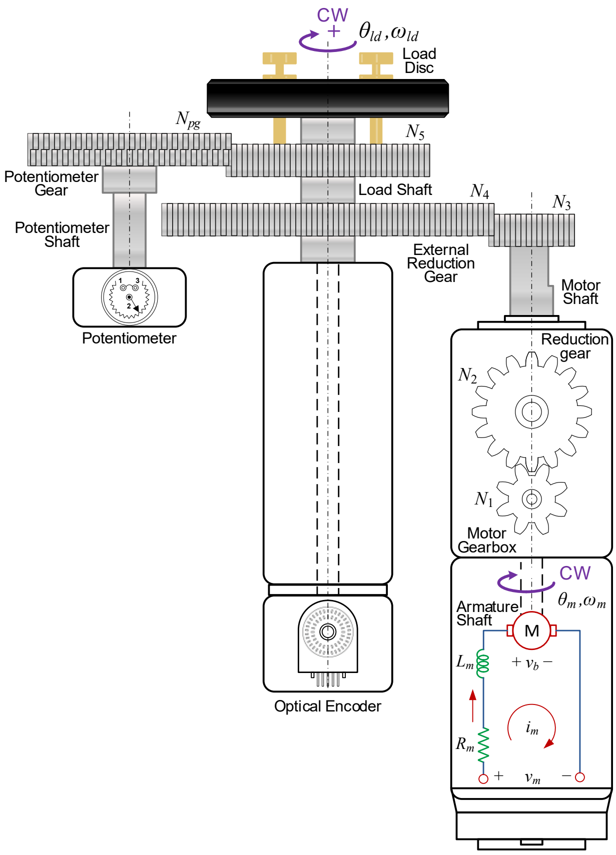


Fig. 2-1. Quanser Rotary Servo Base Unit SRV02 schematic with a load disc (drawn by Dr Arash Khatamianfar).

Table 2-1: Quanser SRV02 DC motor parameters with load.

Item	Symbol	Description	Value
DC Motor	R_m	Motor resistance	2.6 Ω
	L_m	Motor inductance	180×10^{-6} H
	K_m	Torque constant	7.7×10^{-3} N·m/A
	K_b	Motor back-emf constant	7.7×10^{-3} V/(rad/s)
	J_a	Motor inertia	4×10^{-7} kg·m2
	D_a	Motor viscous damping coefficient	8.1×10^{-7} N.m/(rad/s)
External Inertias	J_{g3}	N_3 gear inertia	1×10^{-7} kg·m2
	J_{g4}	N_4 gear inertia	4.2×10^{-5} kg·m2
	J_{g5}	N_5 gear inertia	5.4×10^{-6} kg·m2
	J_{pg}	N_{pg} gear inertia	5.4×10^{-6} kg·m2
	J_{ld}	Load disc inertia	5×10^{-5} kg·m2
	J_{meq}	Total equivalent inertia on motor side	To be calculated by students!
External Viscous Damping Coefficients	D_{g3}, D_{g4}	Main gear system viscous damping	Incorporated in B_{meq}
	D_{g5}, D_{pg}	Potentiometer/anti-backlash gear system viscous damping	Incorporated in B_{meq}
	B_{meq}	Total equivalent viscous damping on motor side	1.44×10^{-6} N.m/(rad/s)
Gear Ratio	$N_1{:}N_2$	Internal planetary great ratio (scaled)	1:14 ($N_1 = 1, N_2 = 14$)
	$N_3{:}N_4$	Main gear ratio (scaled)	1:5 ($N_3 = 1, N_4 = 5$)
	$N_5{:}N_{pg}$	Potentiometer/anti-backlash gear ratio	1:1 ($N_5 = N_{pg}$)

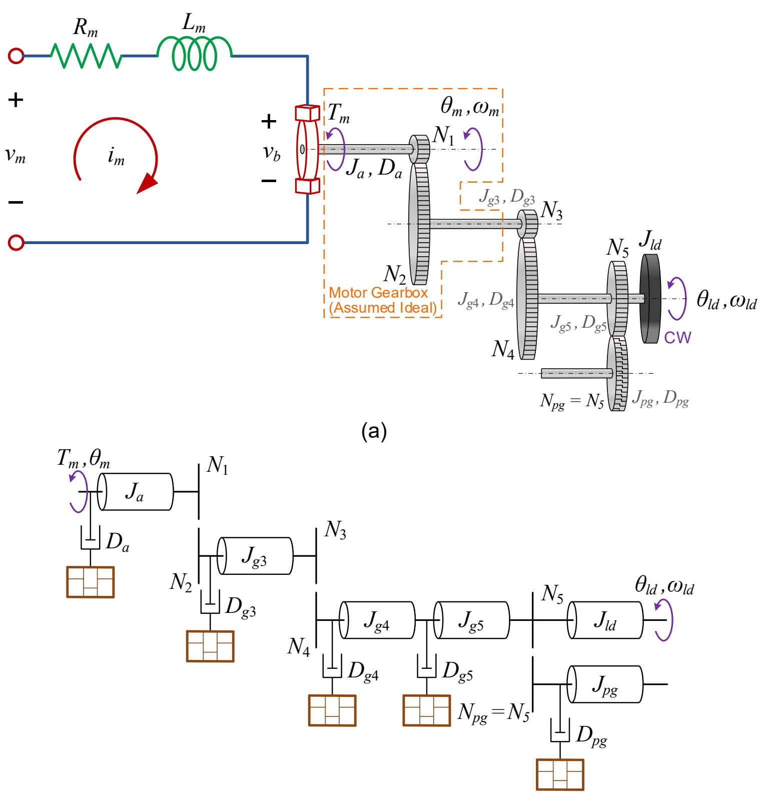


Fig. 2-2. (a) Simplified schematic of SRV02 (b) Diagram of SRV02 mechanical impedances (drawn by Dr. Arash Khatamianfar).

2. **Explain** the conditions under which the transfer functions in Eq. 2-1(a) and Eq. 2-1(b) can be approximated by the lower order forms given in Eq. 2-2(a) and Eq. 2-2(b), respectively. **Show** all working.

$$\frac{\theta_{ld}(s)}{V_m(s)} = \frac{\left(\frac{N_1 N_3}{N_2 N_4}\right)\frac{K_m}{R_m}}{s\left(J_{meq}s + \left(B_{meq} + \frac{K_m K_b}{R_m}\right)\right)}, \quad \text{(a)}$$

$$\frac{\omega_{ld}(s)}{V_m(s)} = \frac{\left(\frac{N_1 N_3}{N_2 N_4}\right)\frac{K_m}{R_m}}{J_{meq}s + \left(B_{meq} + \frac{K_m K_b}{R_m}\right)}. \quad \text{(b)}$$

Eq. 2-2

3. **Rewrite** the approximated DC motor speed transfer function from Eq. 2-2(b) in terms of time constant τ and DC gain K_{DC}, as in Eq. 2-3, by **finding** τ and K_{DC} in terms of the motor parameters in Eq. 2-2(b).

$$\frac{\omega_{ld}(s)}{V_m(s)} = \frac{K_{DC}}{\tau s + 1} \quad \text{Eq. 2-3}$$

⚠ *For the coming questions below, the following MATLAB commands may be useful*: `tf`, `zpk`, `s = tf('s')`, `minreal`, `vpa`, `pole`, `zero`, `dcgain`, `step`, `plot`, `hold on`, `hold off`, `title`, `grid on`, `xlabel`, `ylabel`, `legend`.

4. **Create** the transfer functions in Eq. 2-1 and Eq. 2-2 in MATLAB using the numerical values in Table 2-1 (also available in ELEC3114_Lab2.m file).
   a. **Find** the poles, zeros, and DC gains for each transfer function. ***Hint:*** Refer to the in-class lecture explanation on infinite (or numerically super large) DC gain for motor position transfer function.
   b. **Use** part (**a**) results to **verify** the model order reduction (approximation) for speed transfer function from 2nd-order (Eq. 2-1(b)) to 1st-order (Eq. 2-2(b)).

5. Using the transfer function models in Eq. 2-1(b) and Eq. 2-2(b), **simulate** and **plot** the DC motor **speed response** of this, in **rpm**, to a 4 V step input voltage, as shown in Fig. 2-3. Apply a scaled step input as the motor voltage V_m, and plot the load disc angular velocity (speed) ω_{ld} over time.

⚠ *Hint: Since `step` has a limited plotting functionality, you can save the output from the `step` command and use the `plot` command instead:*

```
[y1,t1]=step(4*Gm_vel,0.15);   % A step-size of 4 units representing 3 volts.
[y2,t2]=step(4*Gm_vel_1stOrder,0.15);
y1_RPM = y1*30/pi;
y2_RPM = y2*30/pi;
figure
subplot(2,1,1)
plot(t1,y1_RPM,'r',"LineWidth",1)
hold on
plot(t2,y2_RPM,'bo')
grid on
xlabel('Time (sec)')
ylabel('Disc Speed (rad/s)')
title('Speed Response of DC motor')
ylim([0 100])
legend('Original 2nd-order','Approximated 1st-order')
```

where `Gm_vel` *is the original 2nd-order model of DC motor speed in Eq. 2-1(b) and* `Gm_vel_1stOrder` *is the 1st-order approximation of the DC motor speed transfer function, for instance.*

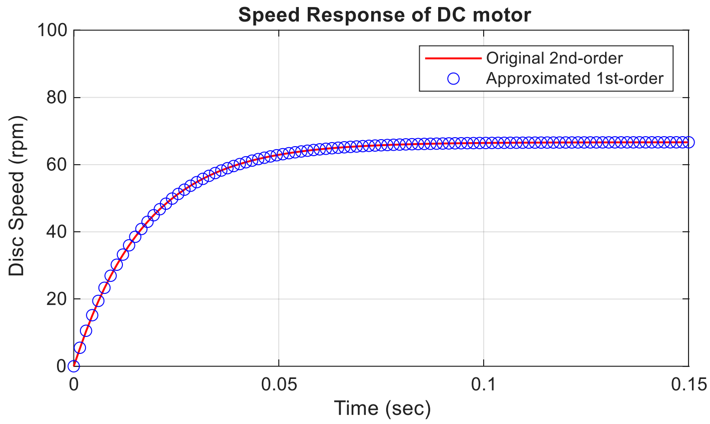


Fig. 2-3. Speed responses of the DC motor transfer function models to an input voltage step of 4V.

6. **Apply** the **Final Value Theorem** to calculate the **steady-state** disc load speed $\omega_{ld_{ss}}$ (in rpm) using either of transfer function models for an input voltage of 4V (from Question 5).
   a. **Compare** this analytical result with the actual steady-state value from your plotted response. Comment on their closeness.

7. **Identify** the pole in the original 2nd-order model (Eq. 2-1(b)) that has the most **dominant effect** on the speed response.
   a. **Justify** whether the 1st-order DC motor speed model is a good approximation of the original model based on pole positions and visually comparing their step responses Fig. 2-3.

8. **State** the **two main conditions** required to justify **approximating** a higher-order model with a lower-order (reduced-order) model.

Lab 2 Exercises

Dynamic Model Derivation (System Identification) for DC Motor

Please complete the following tasks for this lab exercise and **properly document all of your answers and working using your lab book or any other digital notes**.

This exercise aims to **identify** a **transfer function** model for a DC motor using experimental input-output data. Finally you will use your identified model as a basis for an open-loop control system, which reveals the challenges of model uncertainty in control systems without having feedback loop. The experiment uses the Quanser Rotary Servo Base Unit SRV02 (Fig. 2-4), with a load disc mounted on the gear system. The system's electrical and mechanical parameters are provided in Table 2-1 of the Pre-Lab 2 section (also in the ELEC3114_Lab2.m file and MATLAB Grader). The moment of inertia for a solid disc with mass m and radius r is equal to $J = \frac{1}{2}mr^2$.

Fig. 2-4. Quanser Rotary Servo Base Unit SRV02 with a load disc mounted on top.

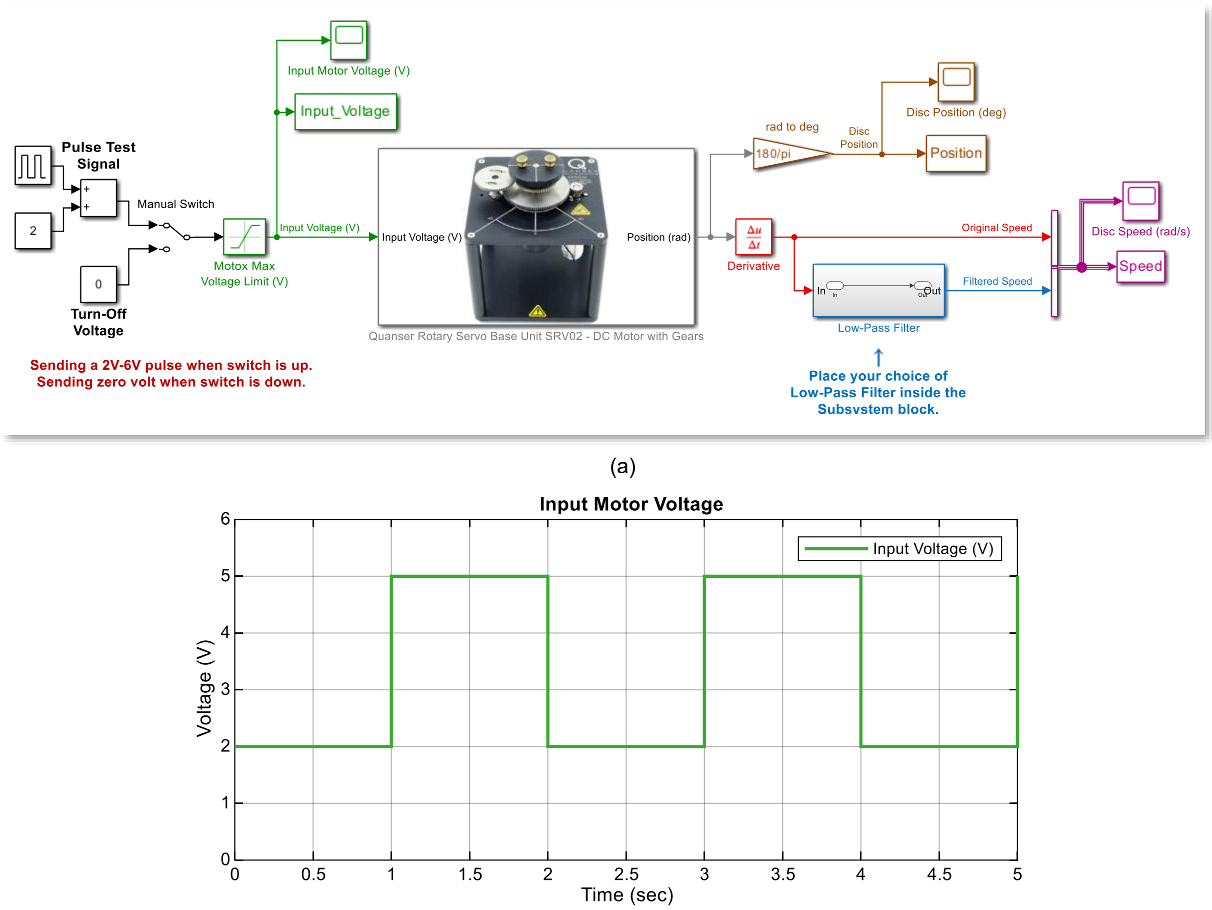
Data Collection – Bump Test

Bump test method is used to identify a **first-order transfer function** model for the Quanser DC motor. The motor's supply voltage can go up to ± 15 V DC, but the data acquisition (DAQ) block **limits** the input voltage to ± 10 V DC. The **nominal (rated or operating) voltage** for this motor is $\pm\mathbf{6}$ **V DC** (think about the difference between nominal and maximum voltages). The maximum allowable **continuous current** is $\pm\mathbf{1}$ **A DC**. To **protect** the device from potential damage you should always use a **Saturation** block in your simulation models to ensure realistic protection and behaviour, specially when this safe voltage limits are exceeded.

For this experiment, Dr. Arash Khatamianfar developed a template Simulink model for real-time operation. You only need to complete and modify the model based on the tasks in this experiment.

1. **Download** the **ELEC3114_Lab2_QuanserDCMotorSRV02_Template_2026.slx** file.
   a. **Open** the template model (Fig. 2-5(a)).
   b. The **Pulse Test Signal** block is configured to generate a **2V-5V** input voltage pulse signal (Fig. 2-5(b)).
   c. The **Solver** for this Simulink model is set to **Fixed-step ode1 (Euler)** with a **sample time** of **2ms (500Hz)**, identical to Pe-Lab 1 written manual (Question 5) with the new speed data. However, the **dominant time constant** of the motor speed response is **much slower** than this sampling rate, so we can assume the response is obtained in **continuous time**.
   d. An **optical encoder**[1] measures the **angular position** of the shaft (see under the Quanser Rotary Servo Base Unit SRV02 – DC Motor block in Fig. 2-5(a)).
   e. To estimate the **disc speed** from the position measurement, a **Derivative** block is used. Note that the Derivate block runs in discrete time. **Right-click** on the block and select **Help** for detailed information on how the block performs **numerical differentiation**.

[1] For more information on encoders, special quadrature optical encoders, you may wish to visit the following websites:
- https://eltra-encoder.eu/news/quadrature-encoder
- https://www.analogictips.com/rotary-encoders-part-1-optical-encoders/#:~:text=A%3A%20The%20encoder%20uses%20the,and%20they%20are%20detected%20separately

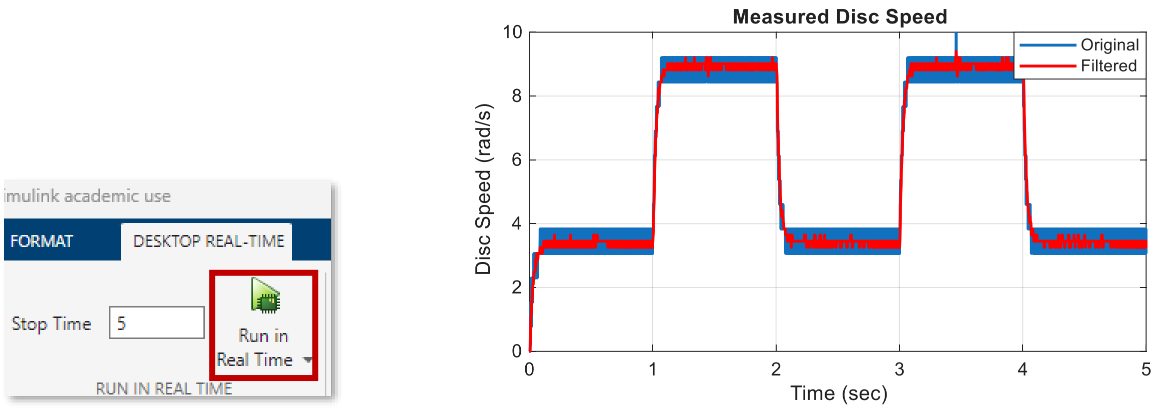


(a)

(b)

Fig. 2-5. Quanser SRV02 DC Motor Template file in Simulink, (a) Block Diagram, (b) Applied pulse test signal for bump test.

2. Make sure the device is **turned on** from the power wall. Run the model by clicking **Run in Real Time** under the Desktop REAL-TIME tab (Fig. 2-6(a)). The **Disk Speed Scope** will pop up, showing the real-time speed in rad/s alongside the pulse test signal via **Input Motor Voltage Scope**.
3. The **disc position** measured via the optical encoder signal is a **staircase-type signal** due to pulse counting for position estimation (you may zoom in on the Disc Position Scope plot). When this signal passes through the Derivative block, it generates noisy spikes. **Explain** one theoretical reason behind this noisy speed signal.
4. Use your learning from Pre-Lab 1 to select a **low-pass filter** (**LPF**) with a suitable **cut-off frequency** and place it inside the **Low-Pass Filter Subsystem** (you may wish to modify the model slightly to follow the Derivative Help example on improving the linearisation of the Derivative block using a transfer function block while filtering noise at the same time).
   a. If a **moving average** filter is chosen as the LPF, **explain** how this method filters noisy data (you may explain it in the time domain or frequency domain).
   b. If a **1st-order transfer function** is chosen as the LPF, **explain** how this method filters noisy data (you may explain it in the time domain or frequency domain).
5. After **adding** your selected **LPF** in the model, **run** the model and show that your filtering approach reduces measurement noise without over smoothing the data, similar to Fig. 2-6(b).
6. **Identify** the necessary condition for the **cut-off frequency** or **time constant** of the filter dynamics (i.e., the speed of the filter) relative to the process dynamics (in this case the DC motor speed dynamics), such that it can be approximated by a **unity gain** and does **not** distort the measurement to the extend where **crucial dynamic information** is **lost** form the collected response.



(a) (b)

Fig. 2-6. Bump test result showing both original unfiltered and filtered DC motor speed data.

Step Response Modelling using Bump Test

In this section, you will prepare the logged bump test data in the MATLAB Workspace to identify the DC motor's dynamic model, including its time constant and DC gain.

Important Notes:

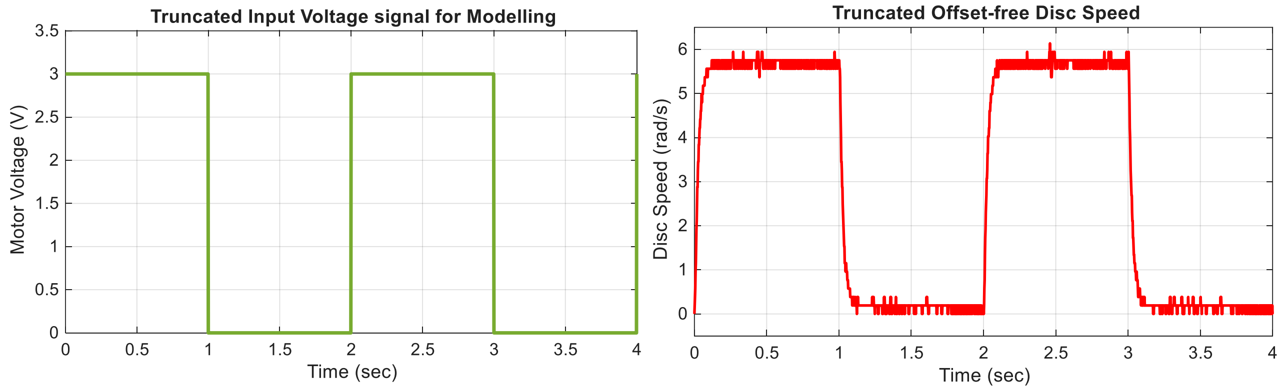
- To save time, use your code from Pre-Lab 1 Question 5, with minor modifications for this section.
- Motor speed data is logged in the MATLAB Workspace as `Speed` (in Structure with Time format).
- **Do NOT** convert the speed data to rpm. **Use** the original **rad/s** speed data for DC motor transfer function identification (think about why that is usually the case).

7. **Remove** the **initial transient** and **DC offset** from your speed and voltage data (as you did in Question 5 of the Pre-Lab 1 manual), then **plot** the results in **MATLAB** as shown in Fig. 2-7(a) and Fig. 2-7(b).

8. Use the **Step Response Modelling** method to **determine** the **DC gain** K_{DC_Iden} and **time constant** τ_{Iden} for the **1st-order DC motor transfer function model**, as shown in Fig. 2-8.

   a. **Create** the 1st-order transfer function in MATLAB.

   b. **Explain** why bump test data provides a more accurate model than using a single step response.



(a) (b)

Fig. 2-7. Measured disc speed data from Simulink model, (a) Offset-free disc speed to be used in step response modelling, (b), Offset-free input voltage to be used for step response modelling.

***Hint**: To estimate the <u>steady-state value</u> (final value) of the speed measurement ω_{ld_ss}, the method used in the Pre-Lab 1 will be useful. You can use either the first or the second transient to find the final value and time constant, as shown in Fig. 2-8. A is the **amplitude** of the **step input signal** (this is the voltage amplitude in Fig. 2-7(a), or step voltage change in Fig. 2-5(b)).*

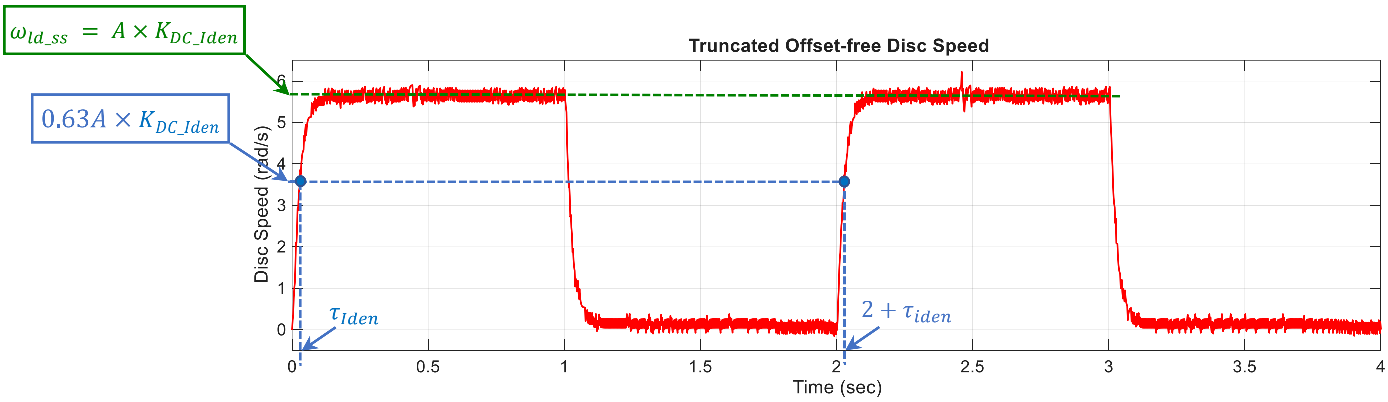


Fig. 2-8. Finding proper DC gain and time constant for a 1st-order model from step response.

Check Point 1 of 3: Please put your name on the **Marking Form**. You may then read the next part of the manual and continue with the next questions until a lab demonstrator is available.

Model Validation

To validate the **identified (experimental)** and **1st-order theoretical (analytical)** DC motor models, compare the **motor's actual response** with the **simulated responses** from both **identified** and **theoretical** transfer function models. Note that this comparison is valid only when the same input voltage is applied to both the real system and the models.

9. **Modify** the Simulink model template to include both the **identified** and **theoretical** transfer function models (Fig. 2-9).

   a. Use the DC motor's **physical parameters** from Table 2-1 in Pre-Lab 2 (also in the **ELEC3114_Lab2.m** file) for the **1st-order theoretical model** in Eq. 2-3 (run the file to load them into MATLAB Workspace).

   b. **Run** the model and **compare** the accuracy of the responses.

   c. **Identify** which model best represents the DC motor's behaviour and dynamics and **explain** why one of the models is relatively **inaccurate**.

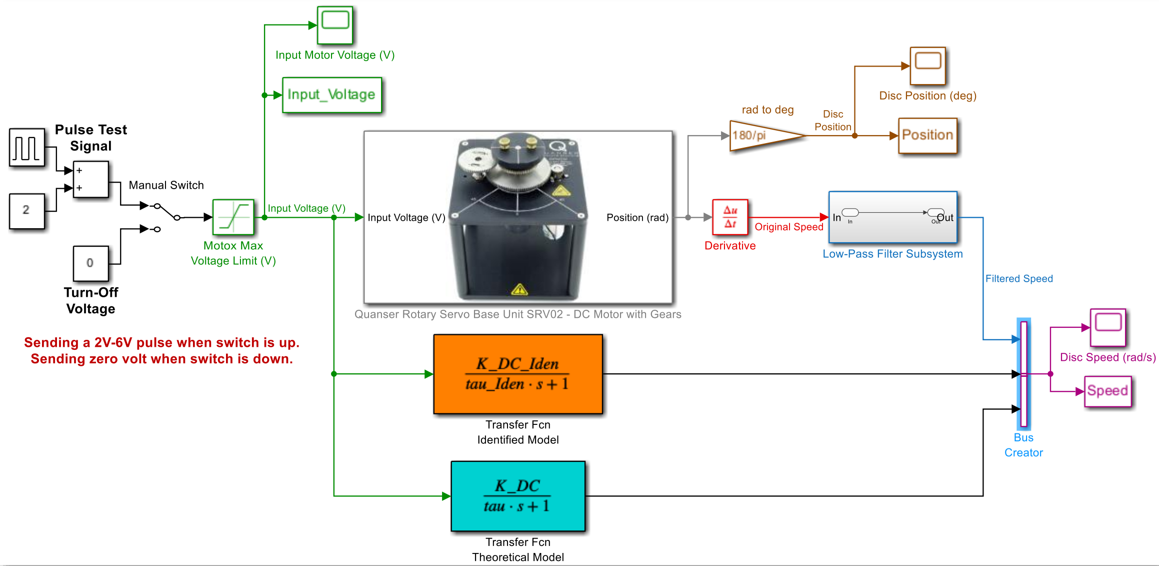


Fig. 2-9. Comparison of the actual filtered DC motor speed response with the identified and theoretical model responses.

10. **Use** the logged data in Workspace and **extract simulated speed data** from the **actual filtered one.**

   a. **Use** `Y_act` to store the actual filtered speed data, `Y_sim_Iden` for the simulated data from the **identified** model, and `Y_sim_Theo` for the simulated data from the **theoretical** model.

   b. **Apply** the **Fit Formula** (Eq. 2-4) to find **two fit values**: one comparing the identified model response with the actual filtered response, and the other comparing theoretical model response with the actual filtered response.

Note: The Fit Formula **quantitatively measures how closely** simulated responses match the actual response, yielding values from 0% to 100%, with **higher percentage** indicating a **better 'fit'**. However, it can be **misleading** when steady-state data dominates transient data, so **visual plots** comparing responses should also be used for **qualitative verification** of model accuracy.

$$Fit = \left[1 - \frac{\|Y_{act} - Y_{sim}\|_2}{\|Y_{act} - \text{mean}(Y_{act})\|_2}\right] \times 100 \qquad \text{Eq. 2-4}$$

Here Y_{act} is the **actual filtered response** and Y_{sim} is the **simulated response** from a transfer function model. $\|X\|_2$ is the **2-Norm (Euclidean Norm)** of a vector X with size n, i.e., $\|X\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots x_n^2}$. You can search online or inside MATLAB Help to find a function that computes the 2-norm of a vector.

11. Based on the fit value, which model is more accurate?

> **Check Point 2 of 3:** Please put your name on the **Marking Form**. You may then read the next part of the manual and continue with the next questions until a lab demonstrator is available.
> Marking will be **closed 15min** before the lab ends.

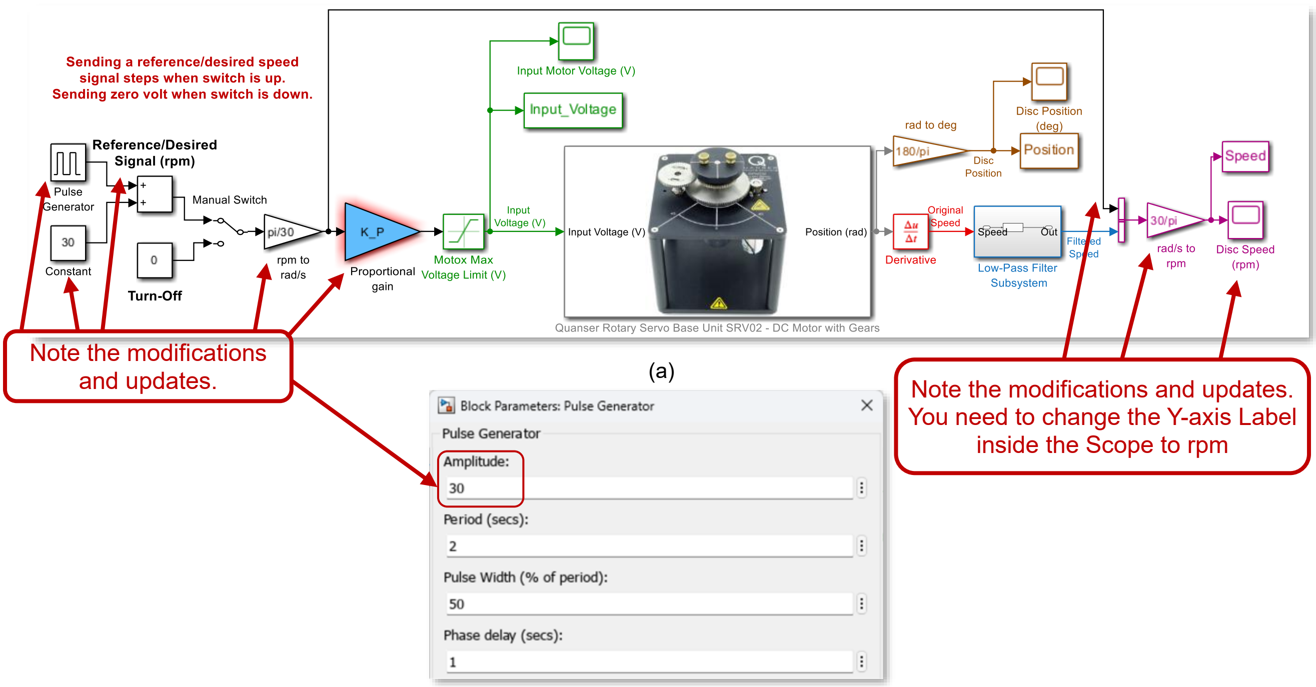
Open-Loop Control

In this checkpoint, we aim to calculate a proportional gain K_P (use `K_P` in MATLAB) such that we can set any reference rotational speed and the motor and its disc spins with the same desired speed.

12. **Modify** the Simulink model template to include a **Gain block** with parameter K_P (to be calculated) and **repurpose** the Pulse Test signal blocks to generate the **reference** motor speed signal (**desired output** or **set-point**) for the open-loop control system (Fig. 2-10(a)).

a. **Update** the Pulse Test Signal blocks in the Simulink Template for a reference (desired) motor speed step levels of **30 rpm (≈ 3.1416 rad/s)** and **60 rpm (≈ 6.283 rad/s)** at intervals of **2 sec** with **50% duty cycle** and a starting time of **1 sec** (Fig. 2-10(b) and the updated constant block in Fig. 2-10(a)). We wish to **display** the response and the reference in **rpm**.

Note: You can **modify text** in the Simulink model by double-clicking text boxes. **Use** proper **conversion** between **rpm** to **rad/s** where needed, as the **controller** requires a **rad/s reference signal** (especially when K_P is based on a model identified using rad/s speed data). For control system **performance** evaluation, **plot** both the **output response** and the reference signal.



(a)

(b)

Fig. 2-10. Modified Simulink model for DC motor open-loop control, (a) Block Diagram model, (b) Pulse Generator block parameters.

13. Use the result from **Tutorial 1, Question 7** to calculate the open-loop **proportional gain** K_P for any constant steady-state reference speed, using the **identified model** from Check Point 1.

a. **Does** K_p need to be updated if the reference speed changes? **Explain** your answer.

b. **Assign** the calculated K_P to the variable `K_P` in MATLAB, then **run** the model to verify that the motor speed **follows** both reference speeds.

c. Is there a **discrepancy** between the steady-state output speed and the reference signal? If so, **explain** its cause and **suggest** a method to correct it.

d. **Set** the Simulink model **Stop Time** to `Inf`. **Run** the model and gradually **adjust** K_P until the steady-state speed **closely** matches the reference speed.

Note: A simple approach is to **replace** `K_P` in the Proportional Gain block with its **numerical** value **multiplied** by **1** (e.g., `0.5*1`). The multiplier acts as a **percentage adjustment**. For example, to reduce K_P by 5% while the device is running, set the multiplier to 0.95 (e.g., `0.5*0.95`), or to increase it by 2.5% set the multiplier to 1.05 (e.g., `0.5*0.95`). Alternatively, use the **Slider Gain** block in Simulink for easier real-time tuning.

e. **Use** that gain to **recalculate** the DC gain of your DC motor K_{DC}.

f. **Explain** why it is **impossible** to achieve accurate speed control in steady state using **open-loop** control.

Check Point 3 of 3: Please put your name on the marking list to get marked before proceeding.

--- End of Lab Experiment 2 ---